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MILITARY SPECIFICATION

FIN AND NOZZLE ASSEMBLY FOR
ROCKET MOTOR 2.75 INCH MARK 4 AND MARK 40
PARTS FOR

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for two types of nozzle and fin assembly designated as Fin and Nozzle Assembly For Rocket Motor 2.75 Inch Mark 4 and Mark 40.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71 -Pallet, Materials Handling
PPP-T-45 -Tape, Gummed, Reinforced and Plain, For Sealing and Securing.
PPP-B-636 -Box Fiberboard

MILITARY

MIL-D-1000 -Drawing, Engineering and Associated Lists.
MIL-A-2550 -Ammunition and Special Weapons, General Specification For.
MIL-I-6866 -Inspection, Penetrant Method Of.
MIL-I-6866 -Inspection Process, Magnetic Particles.

STANDARDS

FEDERAL

Federal Test Method STD No. 151

FSC: 1340

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-151 - Metals, Test Methods.
MIL-STD-1167 - Ammunition Data Card.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for shipment of Inert Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for inspection by Attributes.

DRAWINGS

BUREAU OF ORDNANCE

1517508 - Nozzle & Fin Assembly.

PUBLICATIONS

ET-1517508 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with drawings and specifications.

3.2 Nozzle and Fin Assembly (dwg. 1517508). - The nozzle and fin assembly shall comply with all requirements specified on drawing 1517508 and with all requirements specified in specifications.

3.3 Metal Parts and Other Material. - The metal parts (except the Nozzle Plate which shall meet the requirements of 3.4) and other material used shall be free from defects, including all surface imperfections and discontinuities and all internal defects in the finished component resulting from flaws or lack of soundness in the materials used.

3.3.1 Surface Imperfections. - Surface imperfections include pits, seams, laps, tears, cracks, scale, blisters, cold shuts, and other defects of similar type.

3.3.2 Internal Defects. - Internal defects include pipes,

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lamination, segregations, internal cracks, porosity, flakes, inclusions, and other defects of similar type.

3.4 Magnetic Particle Inspection (dwg. 656925). - When the Nozzle Plate is subjected to magnetic particle inspection, the following requirements shall be met.

3.4.1 Inclusions of sulphides or other ferrous compounds precipitated and rolled into the material, of any degree in the thin section between the counterbored holes or adjacent to the central hole are not acceptable and the part shall be rejected. Shallow discontinuities in other areas of this surface of these holes are acceptable where they are backed up by a 3/16 inch minimum thickness of sound material (see Figure 1).

3.4.2 Inclusions of sulphides or other ferrous compounds precipitated and rolled into the material on either the inside or outside surface of the piston cylinder are unacceptable (see Figure 1).

3.4.3 Inclusions of sulphides or other ferrous compounds precipitated and rolled into the material in the skirt that show on both the inside and outside surfaces are not acceptable, and the part shsll be rejected. Shallow discontinuities in either surface that do not show on the other surface are acceptable (see Figures 1 and 2).

3.4.4 Cracks or cracklike indications are not allowed except in the following areas:

a. On the surfaces adjacent to the detent groove not to exceed 1/16" in depth.

b. On the outside surface of the skirt that does not extend down into the grooves more than 1/16". No cracks are allowed on the "R" surface.

3.5 Pivot Pin Load Test. - The pivot pin shall withstand a static load of twenty pounds applied successively to each end without relative movement between the Spring Pin and the Nozzle Plate.

3.6 Insulation Integrity. - The Nozzle and Fin Assembly prior to installation of Shorting Clip must measure ten (10) megohm minimum resistance when measured with one hundred (100) volts between the Contact Disc and the bourrelet on the Nozzle Plate.

3.7 Burst Strength. - The Burst Diaphragm (dwg. 9220795), when in the configuration of the Nozzle Plate and Insert Assembly (dwg. 1517510) shall rupture at two hundred fifty (250) pounds

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per square inch gage plus or minus seventy five (75) pounds per square inch gage (psig). Pressurization by a fluid shall be at a uniform rate between thirty (30) psig/sec and sixty (60) psig/sec from zero (0) toward three hundred twenty five (325) psig. There shall be no solid material separated from the assembly upon rupture.

3.8 Lead Pull Strength. - The Nozzle and Fin Assembly shall be capable of withstanding a twelve (12) pound pull for two (2) seconds between the Nozzle Plate and the forward connector of the Ground Lead and between the Nozzle Plate and the forward connector of the Live Lead, and between the Nozzle Plate, (with the Cross Head restrained from rearward movement), and the Contact Disc with the Fin Retainer disengaged from the Fin Blades.

3.9 Piston Pressurization Strength. - The Piston Assembly shall be capable of withstanding a pressure of two thousand (2000) pounds per square inch gage (psig), plus fifty (50) psig, minus zero (0) psig applied to that portion of the assembly forward of the "O" Ring groove in the piston. Each piston assembly shall be capable of withstanding the above pressure applied in the above manner for two (2) seconds without pressure drop or without leakage either at the end of the piston between the piston body and the insulation or between the insulation and the conductor.

3.10 Ground Lead Continuity. - The Nozzle and Fin Assembly shall not exceed a resistance of 0.10 ohms between the Nozzle Plate and the forward end of the Ground Lead.

3.11 Air Tightness. - The Nozzle and Fin Assembly shall withstand without a leakage rate of 10-4 cc/sec or greater, an air pressure of 10 psig applied to the inner part of the assembly. The emission of visible air bubbles using approved equipment for submersion in water shall be considered to indicate 10-4 cc leakage rate.

3.12 Workmanship. - All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, metal shavings, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing bare material. All required markings shall be neat and sharply defined. Known item defects (to include complete compliance with 3.1, 3.2 and 3.3) shall not be submitted with the lot.

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3.13 First Article Inspection.-This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement the the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements when such material is controlled by Government or commerical specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Sample

4.2.1 Submission.-The contractor shall submit a First Article

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Sample to a Government approved facility as designated by the Contracting Officer (see para. 6.4) for evaluation in accordance with the provisions of 4.2.2. The First Article Sample shall consist of 30 assemblies and 30 complete sets of parts (i.e., 30 each of every component, and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of fidings, certificates of conformance for materials (see 4.1.1). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A First Article Sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the requirements and inspections, the First Article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and/or submit a new First Article Sample or portion thereof. Until a First Article Sample is accepted the contractor is, in no way authorized by the Government to initiate regular production unless directed by the Contracting Officer.

4.3 Inspection Provisions for Production.

4.3.1 Lot Formation.-The term "inspection lot" as used throughout this specification is defined as an essentially homogeneous collection of units or product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity

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shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than seven thousand (7,000) nor more than fifty thousand (50,000) units.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies in the applicable paragraph headings. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply the AQL's to individual defects using an AQL of 0.65 percent for each major defect and an AQL of 2.50 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.3.2.1 Contact Disc (see dwg. 1253129 covering a detail of dwg. 1517508).

Categories	Defects	Method Of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Plating missing or inadequate.	Visual	01001
Minor:	AQL 2.5 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	01002

4.3.2.2 Fin Retainer (see dwg. 1253131 covering a detail of dwg. 1517508).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	02001

4.3.2.3 Fin Blade (see dwg. 656926 covering a detail of dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.5 percent		
101.	Blade straightness (Note 6A)	Gage	03001
102.	Twist (note 6B)	Gage	03002
103.	Blade symmetry (Note 6C)	Visual	03003
104.	Protective finish missing or improper	Visual	03004
105.	Thickness of blade at hole, boss area	Gage	03005
106.	Crosshead bearing surface not coincident with centerline of hole	Gage	03006
107.	Wall thickness around hole, min	Gage	03007
108.	Distance from center of hole to leading edge	Gage	03008
Minor:	AQL 4.0 percent		
201.	Height (Trailing edge to leading edge)	Gage	03009
202.	Radius of material around hole	Gage	03010
203.	Centerline of hole perpendicular to boss	Gage	03011
204.	Distance between crosshead bearing corner and center of hole	Gage	03012
205.	Length from center of hole to slotted end	Gage	03013
206.	Thickness at slotted end	Gage	03014
207.	Height of boss from crosshead bearing corner	Gage	03015

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208.	Distance from boss at leading edge of boss to leading edge of blade (does not apply to fin blades made to detail "R" optional construction or flat alternative construction where leading edge of boss and leading edge of blade coincide)	Gage	03016
	or		
208.	Distance from boss at leading edge of boss to center of hole (applies to Fin Blades made to detail "R" optional construction only)	Gage	03017
209.	Evidence of poor workmanship (see 3.12)	Visual	03018

4.3.2.4 Spring Pin (see dwg. 9241782 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	04001
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	04002

4.3.2.5 Cross Head (see dwg. 456909 covering a detail of dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Both faces not perpendicular to piston hole	Gage	05001
102.	Concentricity of piston hole to outside diameter	Gage	05002
103.	Protective finish missing or improper	Visual	05003

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Minor:	AQL 4.0 percent		
201.	Diameter of crosshead, max....	Gage	05004
202.	Thickness.....	Gage	05005
203.	Width of cross bars.....	Gage	05006
204.	Evidence of poor workmanship (see 3.12).....	Visual	05007

4.3.2.6 Nut, Self-Locking (see dwg. MS21044-N3 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	06001

4.3.2.7 "O" Ring (see dwg. 650950 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Nicks, cuts, surface defects or surface irregularities.....	Visual	07001
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	07002

4.3.2.8 "O" Ring (see dwg. 650953 covering a detail of dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Diameter of cross section.....	Gage	08001
102.	Nicks, cuts, surface defects or surface irregularities.....	Visual	08002

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	08003

4.3.2.9 Piston (Glass Bead One End) (see dwg. 9220796 covering a detail of dwg. 1517508). This paragraph shall be applied when Piston Assembly (Glass Bead, One End) dwg. 9220782 is supplied in the Nozzle and Fin Assembly.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Diameter of knurled section after knurling	Gage	09001
102.	Plating missing or improper	Visual	09002
103.	Head diameter	Gage	09003
104.	Body diameter	Gage	09004
105.	Cross head shaft diameter	Gage	09005
106.	Thread improper	Gage	09006
107.	Concentricity of head diameter with body diameter	Gage	09007
Minor:	AQL 2.50 percent		
201.	Length of knurl	Gage	09008
202.	Length of body	Gage	09009
203.	Evidence of poor workmanship (see 3.12)	Visual	09010

4.3.2.10 Piston Assembly (Glass Bead, One End), (see dwg. 9220782 covering a detail of dwg. 1517508). This paragraph shall be applied when this assembly is supplied in the Nozzle and Fin Assembly.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Glass bead cracked, loose or missing	Visual/ Manual	10001
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	10002

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4.3.2.11 Piston (Glass Bead, Two Ends), (see dwg. 9220783 covering a detail of 1517508). This paragraph shall be applied when Piston Assembly (Glass Bead, Two Ends) dwg. 9220797 is supplied in the Nozzle and Fin Assembly.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Diameter of knurled section after knurling	Gage	11001
102.	Head diameter	Gage	11002
103.	Body diameter	Gage	11003
104.	Cross head shaft diameter	Gage	11004
105.	Thread improper	Gage	11005
106.	Concentricity of head diameter with body diameter	Gage	11006
Minor:	AQL 4.0 percent		
201.	Length of knurl	Gage	11007
202.	Length of body	Gage	11008
203.	Evidence of poor workmanship (see 3.12)	Visual	11009

4.3.2.12 Piston Assembly (Glass Bead, Two Ends), (see dwg. 9220797 covering a detail of dwg. 1517508). This paragraph shall be applied when this assembly is supplied in the Nozzle and Fin Assembly.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Glass bead cracked, loose or missing	Visual/ Manual	12001
102.	Plating missing or improper	Visual	12002
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	12003

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4.3.2.13 Electrical Lead Live Assembly. - Prior to assembly of forward connector (see dwg. 9220785 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Overall length of assembly.....	Gage	13001
Minor:	None defined		

4.3.2.14 Electrical Lead Live Assembly (see dwg. 9220785 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Part missing.....	Visual	14001
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	14002

4.3.2.15 Fin retainer & Live Lead Assembly (see dwg. 9234433 covering a detail of dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Overall length of assembly.....	Gage	15001
102.	Contact disc not properly or securely fastened to fin retainer.....	Visual/ Manual	15002
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	15003

4.3.2.16 Electrical Lead Ground before assembly of forward connector (see dwg. 1253133 covering a detail of 1517508).

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Length of Wire, Electrical.....	Gage	16001
Minor: None defined.			

4.3.2.17 Electrical Lead Ground (see dwg. 1253133 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 2.50 percent		
201.	*Evidence of poor workmanship* (see 3.12).....	Visual	17001

4.3.2.18 Nozzle Plate (see dwg. 656925 covering a detail of 1517508).

4.3.2.18.1 Nozzle Plate, before plating (see dwg. 656925 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Diameter of nozzle port, max..	Gage	18001
102.	Concentricity of pitch circle to outside diameter at forward section of plate.....	Gage	18002
Minor:	AQL 4.0 percent		
201.	Diameter of nozzle port counterbore.....	Gage	18003
202.	Depth of nozzle port counterbore.....	Gage	18004
203.	Evidence of poor workmanship (see 3.12).....	Visual	18005

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4.3.2.18.2 Nozzle plate after plating - Inserts Assembled (see dwg. 656925 covering a detail of dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 2.5 percent		
101.	Concentricity "O" ring groove diameter to outside diameter at forward section of plate	Gage	19001
102.	Concentricity of lockwire groove diameter to outside diameter at forward section of plate	Gage	19002
103.	Concentricity of bourrelet to outside diameter at forward section of plate	Gage	19003
104.	Concentricity of piston hole diameter at aft surface to outside diameter at forward section of plate	Gage	19004
105.	Perpendicularity of lockwire groove to axis of outside diameter at forward section of plate	Gage	19005
106.	Nicks, radial scratches, gouges, and spiral tool marks on forward face of plate and in "O" ring groove	Visual	19006
107.	Outside diameter of forward section of plate	Gage	19007
108.	"O" Ring groove diameter	Gage	19008
109.	Location of fin pivot holes from center of bourrelet diameter	Gage	19009
110.	Diameter of fin pivot holes, min	Gage	19010
111.	Width of fin pivot slots, max	Gage	19011
112.	Thickness of material below lockwire hole	Gage	19012
113.	Outside diameter at fin pivot ears, min	Gage	19013
114.	Width of lockwire groove	Gage	19014
115.	Diameter of lockwire groove	Gage	19015
116.	Concentricity of piston hole diameter at aft surface with forward piston hole diameter	Gage	19016
117.	Forward piston hole diameter	Gage	19017
118.	Diameter of piston hole in aft face of plate	Gage	19018

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Minor:		AQL 6.5 percent	
201.	Length from outside edge of fin pivot to mid point between fin pivot slots - typical, max	Gage	19019
202.	Radius of outside chamfer on ears, min	Gage	19020
203.	Lockwire hole off location	Gage	19021
204.	Diameter of lockwire hole,min.	Gage	19022
205.	Length from top of fin pivot ear to "R" surface of plate, min	Gage	19023
206.	Inside diameter, min	Gage	19024
207.	Diameter of detent groove,max.	Gage	19025
208.	Diameter of spot face, min	Gage	19026
209.	Distance between bottom of forward piston hole diameter and spot face	Gage	19027
210.	Length from center of fin pivot hole to aft face of plate, max	Gage	19028
211.	Width of detent slot	Gage	19029
212.	Length from step to rear face of lockwire groove	Gage	19030
213.	Length from forward face to step	Gage	19031
214.	Length from step to front of detent slot	Gage	19032
215.	"O" Ring groove radius or chamfer oversize	Gage	19033
216.	Lockwire groove radius or chamfer oversize	Gage	19034
217.	Concentricity of inside diameter to outside diameter at forward section of plate	Gage	19035
218.	Concentricity of detent groove to outside diameter at forward section of plate	Gage	19036
219.	Forward face parallel to lockwire groove	Gage	19037
220.	Width of "O" ring groove	Gage	19038

4.3.2.19 Nozzle Insert (see dwg. 456916 covering a detail of dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined.

Major:		AQL 1.0 percent	
101.	Concentricity of nozzle throat with nozzle exit diameter.....	Gage	20001
102.	Concentricity of .500 radius with nozzle exit diameter.....	Gage	20002
103.	Throat mismatch to .500 radius.....	Visual	20003
104.	Outside diameter of press fit, min.....	Gage	20004
105.	Concentricity of outside diameter of press fit with nozzle exit diameter.....	Gage	20005
Minor:		AQL 4.00 percent	
201.	Length of shoulder.....	Gage	20006
202.	Length of insert, min.....	Gage	20007
203.	Length of nozzle throat, min..	Gage	20008
204.	Evidence of poor workmanship (see 3.12).....	Visual	20009

4.3.2.20 Nozzle Plate and Insert Assembly before plating and assembly of Burst Diaphragm (see dwg. 1517510 covering a detail of 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 2.50 percent		
101.	Nozzle Insert stop shoulder not fully and firmly seated against nozzle plate.....	Visual	21001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	21002

4.3.2.21 Diaphragm, Burst (Nozzle Insert) (see dwg. 9220795 covering a detail of dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Outside diameter, min.....	Gage	22001
102.	Length of coin, min.....	Gage	22002
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	22003

4.3.2.22 Nozzle Plate and Insert Assembly (see dwg. 1517510 covering a detail of 1517508).

4.3.2.22.1 Nozzle Plate and Insert Assembly (see dwg. 1517510 covering a detail of 1517508). This paragraph shall be applied to Mark 4 Nozzle and Fin Assemblies.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Nozzle throat diameter.....	Gage	23001
102.	Plating missing or improper...	Visual	23002
103.	Burst Diaphragm positioned improperly (to include off center and/or backward) or missing...	Visual	23003
104.	Perpendicularity of nozzle cone to lockwire groove "V" surface.....	Gage	23004
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12).....	Visual	23005

4.3.2.22.2 Nozzle Plate and Insert Assembly (see dwg. 1517510 covering a detail of 1517508). This paragraph shall be applied to Mark 40 Nozzle & Fin Assemblies.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 1.0 percent		
101.	Nozzle Throat diameter	Gage	23001
102.	Plating missing or improper	Visual	23002
103.	Burst diaphragm positioned improperly (to include off center and/or backward) or missing	Visual	23003
104.	Height of forward end of scarf from forward face	Gage	23004
105.	True position of plane of scarf	Gage	23005
106.	Perpendicularity of nozzle cone to lockwire groove "V" surface	Gage	23006
Minor:	AQL 2.50 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	23007

4.3.2.23 Nozzle and Fin Assembly (see dwg. 1517508).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Assembly does not pass through launch tube simulator	Gage	25001
Major:	AQL 1.0 percent		
101.	Part missing	Visual	25002
102.	Fin retainer not securely assembled	Visual/Manual	25003
103.	Contact Disc, connector and lead not properly insulated	Visual	25004
104.	Fin spacing unsymetrical	Visual	25005
105.	Terminal wire bent excessively	Visual	25006
106.	Distance of uninsulated terminal wire exposed on aft end	Gage	25007
Minor:	AQL 2.50 percent		
201.	Marking improper or missing	Visual	25008
202.	Evidence of poor workmanship (see 3.12)	Visual	25009

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4.3.3 Testing

4.3.3.1 Fin Blade Tensile Test - Major Defect. - Each furnace (heat treat) lot of fin blades shall be sampled and tested for fin blade strength. Furnace (heat treat) lots shall be sampled and tested according to the following:

a. According to 4.3.3.1.1, until five consecutive lots have passed fin blade material verification. Thereafter, one lot in five furnace (heat treat) lots shall be randomly selected, and so tested.

b. All furnace (heat treat) lots which are not sampled and tested according to 4.3.3.1.1 shall be sampled and tested according to 4.3.3.1.2.

c. Rejection of any furnace (heat treat) lot shall require return to testing as in "a" above.

4.3.3.1.1 Fin Blade Material Verification Test - Major Defect-Code No. 26001 (see dwg. 656926 Note 1.A (1) or (2). - Three (3) fin blades from each furnace (heat treat) lot to be tested shall be subjected to a tensile pull test as specified in 4.4.1.1 using equipment in accordance with 4.3.4. Failure of two or more fin blades to meet the requirements of Note 1 of dwg. 656926 shall cause furnace (heat treat) lot rejection. In the event that lot rejection criteria are reached due to no test (machine failure, specimen failing outside gage length, etc.) each no test fin blade shall be replaced in the sample by another fin blade drawn from the heat treat lot.

4.3.3.1.2 Fin Blade Pin Pull Test. Major Defect-Code No. 26002 (see dwg. 656926 Note 1.A (3). - Ten (10) fin blades from each furnace (heat treat) lot to be tested shall be subjected to a tensile pull test as specified in 4.4.1.2 using equipment in accordance with 4.3.4. Failure of one or more fin blades to meet the requirements of Note 1.A (3) on dwg. 656926 shall cause furnace (heat treat) lot rejection. Failure of the fin blade to be pulled to the minimum load attributable to the test equipment (such as pin failure) shall be considered a no test fin blade. In the event of more than one no test fin blade in the sample, additional fin blades shall be selected so that nine fin blades shall be validly tested.

4.3.3.2 Hardness Test on Spring Pin (see dwg. 9241782 covering a detail of 1517508) - Major Defect - Code No. 04003. - A sample consisting of twenty-five (25) Spring Pins shall be taken from each lot of Spring Pins. The sample shall be tested as specified in 4.4.2 using equipment in accordance with 4.3.4. Failure of two (2) or more specimens to indicate hardness within the specified limits shall cause rejection of the lot. This test is considered destructive and parts tested shall not be returned to the lot.

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4.3.3.3 Vise Test on Spring Pin (see dwg. 9241782 covering a detail 1f 1517508) - Major defect - Code No. 04004. - A sample consisting of one hundred twenty-five (125) spring pins shall be taken from each lot of spring pins. The sample shall be tested as specified in 4.4.3 using equipment in accordance with 4.3.4 Any pin in which a crack appears shall be considered defective. Three (3) or more defective spring pins in the sample shall cause rejection of the lot. This test is considered destructive and parts tested shall not be returned to the lot.

4.3.3.4 Compression Test on Cross Head (see dwg. 456909 covering a detail of 1517508) - Major Defect - Code No. 05008. This paragraph shall apply when a powdered metal Cross Head is supplied in the Nozzle and Fin Assembly. A sample consisting of fifty (50) Cross Heads shall be taken from each lot of Cross Heads. The sample shall be tested as specified in 4.4.4 using equipment in accordance with 4.3.4. Failure of any Cross Head in the sample to meet the requirements of Note 8.A. on dwg. 456909 shall cause rejection of the lot.

4.3.3.5 Bending Test on Cross Head (see dwg. 456909 covering a detail of 1517508) - Major Defect - Code No. 05009. This paragraph shall apply when a powdered metal Cross Head is supplied in the Nozzle and Fin Assembly. A sample consisting of fifty (50) Cross Heads shall be taken from each lot of Cross Heads. The sample shall be tested as specified in 4.4.5 using equipment in accordance with 4.3.4. Failure of any Cross Head in the sample to meet the requirements of Note 8.B. on dwg. 456909 shall cause rejection of the lot.

4.3.3.6 Pressure Test on Piston Assembly (see dwg. 9220782 or dwg. 9220797 covering alternate details of dwg. 9220785) (see 3.9) - Major Defect - Code No. 10003 (for Piston Assembly (Glass Bead, One End)) or Code No. 12004 (for Piston Assembly (Glass Bead, Two Ends)). This test shall be performed on the Piston Assembly with the Code No. dependant on the drawing to which the supplied assembly conforms. The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4. Any Piston Assembly which does not meet the requirements specified in 3.9 shall be considered defective.

4.3.3.7 Magnetic Particle Inspection of Nozzle Plate (dwg. 656925 covering a detail of 1517510). (see 3.4) - Critical Defect - Code No. 18006. Each lot of Nozzle Plates shall be subjected to one hundred percent magnetic particle inspection as specified in 4.4.7 using equipment in accordance with 4.3.4. Any Nozzle Plate failing to meet the requirements specified in 3.4 shall be considered defective. All defective Nozzle Plates shall be removed from the lot.

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4.3.3.8 Burst Test on Nozzle Plate and Insert Assembly (dwg. 1517510 covering a detail of 1517508) (see 3.7). Major Defect - Code No. 23010. A sample of twenty (20) Nozzle Plate and Insert Assemblies shall be taken from each Nozzle Plate and Insert Assembly lot. The sample shall be tested as specified in 4.4.8 using equipment in accordance with 4.3.4. A Burst Diaphragm which fails to rupture as specified in 3.7 shall be counted as one defect. (Therefore, each Nozzle Plate and Insert Assembly in the sample has a potential of four (4) defects). If there is no defect in the sample, the lot shall be accepted. Three or more defects shall cause rejection of the lot. One or two defects shall cause a second sample consisting of thirty (30) Nozzle Plate and Insert Assemblies to be taken from the same lot. The second sample shall be tested as specified in 4.4.8 using equipment in accordance with 4.3.4. The number of defects in the first sample shall be added to the number of defects in the second sample and a total of three or more defects in the combined samples shall cause rejection of the lot. This test is considered destructive and parts tested shall not be returned to the lot.

4.3.3.9 Piston Free Movement Test on Nozzle and Fin Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blades (see dwg. 1517508). Major Defect - Code No. 24001. - The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL 0.65 percent. This test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4. Any Nozzle & Fin Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blades in which the Piston will not freely move one half (1/2) inch to the rear and return forward shall be considered defective.

NOTE: This test shall be performed in such a manner that the Piston Movement Restriction of Note 2 BUORD dwg. 1517508 is maintained.

4.3.3.10 Fin Blade Free Fall Test on Nozzle and Fin Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blade (see dwg. 1517508). Major Defect - Code No. 24002. - The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.65 percent. This test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4. Any Nozzle and Fin Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blades on which any Fin Blade does not fall of its own weight when tested shall be considered defective.

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4.3.3.11 Fin Pivot Pin Push Test on Nozzle and Fin Assembly (see dwg. 1517508) - Major Defect - Code No. 25010. The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL 0.65 percent. This test shall be performed as specified in 4.4.11 using equipment in accordance with 4.3.4. Any Nozzle and Fin Assembly in which any Spring Pin moves when tested shall be considered defective.

4.3.3.12 Air Test on Nozzle and Fin Assembly (see dwg. 1517508) (see 3.11) - Major Defect - Code No. 24003. The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4. Any Nozzle and Fin Assembly which indicates a leakage rate in excess of the requirement shall be considered defective. This test may be performed prior to complete assembly of all parts provided the Nozzle Plate and Insert Assembly (dwg. 1517510), O-Ring (dwg. 650950), O-ring (dwg. 650953), Crosshead (dwg. 456909), Nut Self-Locking (dwg. MS 21044-N3), and at least the Piston Assembly Lead Live Assembly have been assembled to one another to form the units sampled for test.

NOTE: This test shall be performed in such a manner that the Piston Movement Restriction of Note 2 BUORD dwg. 1517508 is maintained.

4.3.3.13 Lead Pull Test on Nozzle & Fin Assembly, with the Fin Retainer and Contact Disc disengaged from the Fin Blades. (Dwg. 1517508) (see 3.8) - Major Defect - Code No. 24004. The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.13 using equipment in accordance with 4.3.4. If the Electrical Lead Live Assembly or the Ground Lead exhibit separation or relative movement between any of the components, the assembly shall be considered defective.

4.3.3.14 Insulation Integrity Test on Nozzle and Fin Assembly (dwg. 1517508) (see 3.6) - Major Defect - Code No. 25011. - The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. The test shall be performed as specified in 4.4.14 using equipment in accordance with 4.3.4. Any Nozzle and Fin Assembly which indicates less than the required resistance shall be considered defective.

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4.3.3.15 Electrical Continuity Test on Ground Lead of Nozzle and Fin Assembly. (Dwg. 1517508) (see 3.10)-Major Defect-Code No. 25012.-The sample plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.15 using equipment in accordance with 4.3.4. Any Nozzle and Fin Assembly, in which the resistance in the ground circuit is indicated as exceeding that specified shall be considered defective.

4.3.3.16 Electrical Continuity Test on Live Lead of Nozzle and Fin Assembly - (see dwg. 1517508).-Major Defect - Code No. 25013.-The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.16 using equipment in accordance with 4.3.4. Any Nozzle and Fin Assembly, in which the resistance in the live lead is indicated as exceeding that specified in Note 14 on dwg. 1517508 shall be considered defective.

4.3.3.17 Electrical Continuity on Spring Pin (see dwg. 1517508)-Major Defect-Code No. 25014.-The sampling plan for this test shall be in accordance with standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.17 using equipment in accordance with 4.3.4. Any Nozzle and Fin Assembly in which the resistance between the inside diameter of the Spring Pin and the bourrolet is indicated as exceeding that specified in Note 20 on dwg. 1516508 shall be considered defective.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET 1517508 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.4 Test Methods and Procedures

4.4.1 Fin Blade Tensile Test (see 4.3.3.1).

4.4.1.1 Fin Blade Material Verification Test (see 4.3.3.1.1). A specimen shall be prepared from each fin blade in the sample. Each specimen shall be pulled in accordance with Tension Test Method 211.1 of Federal Test Method STD No. 151.

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4.4.1.2 Fin Blade Pin Pull Test (see 4.3.3.1.2). - Each fin blade in the sample shall be subjected to a pull force to the leading edge. The aft portion of the fin blade shall be gripped using suitable jaws which shall not grip any portion of the fin blade two inches or more from the aft end of the fin blade. The application of force shall be by a pin through the hole in the fin blade having a diameter of 0.187 ± .002 inches. The pin shall be supported on both sides of the fin blade with a distance between supports of 0.3715 plus .0010 minus 0 inches. The force shall be applied until failure of the fin blade occurs or until a load force of three thousand pounds (3,000 lbs.) is reached. Observation shall be made of the load at failure or 3,000 lbs. which ever condition is met. Observation shall be made of the load force at failure.

NOTE: Pin hardness of approximately Rockwell 57 is recommended.

4.4.2 Hardness Test on Spring Pin (see 4.3.3.2). - A specimen shall be prepared from each Spring Pin in the sample. Each specimen shall be prepared and tested in accordance with para. 5.2 of AMS 7207 and Federal Test Method STD. No. 151, Method 243.1.

4.4.3 Vise Test on Spring Pin (see 4.3.3.3). - Each Spring Pin in the sample shall be placed in a vise and the gap in the Spring Pin squeezed closed. Each Spring Pin shall be removed and examined for cracks.

4.4.4 Compression Test on Cross Head (see 4.3.3.4). - Each Cross Head in the sample shall be subjected to a compression load of one thousand (1,000) pounds, applied alternately at the outer ends of each set of diametrically opposite arms. Visual observation shall be made for permanent deformation of the Cross Head.

4.4.5 Bending Test on Cross Head (see 4.3.3.5). - Each Cross Head in the sample shall be subjected to a bending load of five hundred (500) pounds applied at the center of the Cross Head by a .217 Inch, Maximum, diameter ram, with the Cross Head supported alternately by supports under each set of diametrically opposite arms. The supports shall be a minimum of .950 inches apart. Visual observation shall be made for permanent deformation of the Cross Head.

4.4.6 Pressure Test on Piston Assembly (see 4.3.3.6). - Each Piston Assembly in the sample shall be held in such a way as to subject that portion of the assembly forward of the "O" ring groove to two thousand (2,000) psig pressure, plus fifty (50) psig, minus zero (0) psig for at least two (2) seconds. Observation shall be made for leakage or pressure drop.

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4.4.7 Magnetic Particle Inspection of Nozzle Plate (see 4.3.3.7). - Each Nozzle Plate shall be magnetic particle inspected in accordance with MIL-I-6868. For surface indications Penetrant Method of Inspection in accordance with MIL-I-6866 may be used.

4.4.8 Burst Test on Nozzle Plate and Insert Assembly (see 4.3.3.8). - The burst test shall be performed on each Burst Diaphragm in each Nozzle Plate and Insert Assembly in the sample. Each Burst Diaphragm shall be individually subjected to a pressure applied to the forward side of the Burst Diaphragm at a rate in accordance with 3.7. Observation shall be made for the pressure at rupture and each Burst Diaphragm shall be examined for the presence of all material.

4.4.9 Piston Free Movement Test (see 4.3.3.9). - Each Nozzle and Fin Assembly in the sample with the Fin Retainer and Contact Disc disengaged from the Fin Blades shall be partially functioned by moving the piston one half (1/2) inch aft. The piston shall then be returned to the forward position. Do not move the piston more than five eighths (5/8) inch. Excessive movement will damage serrations on the piston.

4.4.10 Fin Blade Free Fall Test (see 4.3.3.10). - Each Nozzle and Fin Assembly in the sample, with the Fin Retainer and Contact Disc disengaged from the Fin Blades shall be supported on the "R" surface of the nozzle plate. The Fin Blades shall be raised to a position which is at least sixty (60) degrees above the horizontal. The fins shall be released and allowed to fall of their own weight.

4.4.11 Fin Pivot Pin Push Test (see 4.3.3.11). - Each Nozzle and Fin Assembly in the sample shall be held with the nozzle plate positively positioned. A static load of twenty (20) pounds shall be applied alternately to each end of each spring pin. Observation shall be made for movement during application of the load.

4.4.12 Air Test (see 4.3.3.12). - Each Nozzle & Fin Assembly in the sample shall be subjected to ten (10) psig air pressure internally for at least ten (10) seconds. Observation shall be made for bubbles if a wet submergence test is utilized; however, parts must be thoroughly dried after completion of the test. If a dry method of test is used, observation shall be made for the leakage rate. The crosshead must be restrained in the forward position during testing.

4.4.13 Lead Pull Test (see 4.3.3.13). - Each Nozzle & Fin Assembly in the sample with the Fin Retainer and Contact Disc disengaged from the fin blades shall undergo the following: The nozzle plate shall be held with the crosshead secured from movement. Alternately, the forward connector on the live lead, the contact disc on the live lead, and the forward connector on the ground lead

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shall have a twelve (12) pound pull applied for at least two (2) seconds. Observation shall be made for relative movement between any part of any lead and the nozzle plate.

4.4.14 Insulation Integrity Test - (see 4.3.3.14).-Each Nozzle and Fin Assembly in the sample shall have a one hundred (100) volt potential applied to measure the resistance between the Contact Disc and the bourrelet of the Nozzle Plate.

4.4.15 Electrical Continuity Test on Ground Lead-(see 4.3.3.15). The resistance of each Nozzle and Fin Assembly in the sample shall be measured between the bourrelet on the Nozzle Plate and the forward connector on the Ground Lead.

4.4.16 Electrical Continuity Test on Live Lead-(see 4.3.3.16). The resistance of each Nozzle and Fin Assembly in the sample shall be measured between the Contact Disc and the forward connector on the Electrical Lead Live Assembly.

4.4.17 Electrical Continuity Test on Spring Pin-(see 4.3.3.17). The resistance of each Nozzle and Fin Assembly in the sample shall be measured between the inside diameter of the Spring Pin and the bourrelet of the Nozzle Plate.

5. PREPARATION FOR DELIVERY.

5.1 Packing.-Level C.-Forty (40) nozzle and fin assemblies shall be packed in a PPP-B-636, Type CF, Style CSSC, Class Domestic, Variety SW, Grade 275 fiberboard box. Each box shall be fitted with half-slotted partitions assembled to form 40 individual cells in an 8 by 5 arrangement. Partitions shall be constructed of the same fiberboard as the box except "B" flute, Grade 200 board shall be used. Each cell shall hold one assembly in an upright position. Provision shall be made to protect the forward end of the live lead wire which protrudes out of the pistion. Boxes shall be closed by application of 2 inch wide tape, PPP-T-45, Type III Grade B, tape over the full length of the top and bottom seams of the box with approximately 2 inch minimum extension of tape onto the end panels and around box corners.

5.2 Palletization.-Boxes shall be palletized for shipment. (see 6.5).-Pallets shall conform to NN-P-71, Type II or II, Size 1 or 2. The maximum gross weight per pallet shall be 2,000 pounds. A minimum of 4 straps - 2 straps around the width and length of the pallet load shall secure the load. Strapping shall conform to QQ-S-781, Type I or IV, 5/8 x .020. Provision shall be made to prevent strapping from cutting into the fiber boxes without loss of tension by use of nominal 1 x 4 wooden boards placed under the straps or by use of edge protectors.

5.3 Marking.-Each box and pallet load shall be marked in accordance with paragraph 3.7.2.2, 3.7.4 and 3.7.4.1 of MIL-STD-1169.

6. NOTES

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6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall by submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections

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must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal ATTN: SMUPA-QA-A-R. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First Article Samples.-Instruction as to the location for the evaluation of the First Article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-I.

6.5 It is recommended that re-use of the wooden pallets be authorized.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United

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States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Preparing Activity:
Army-MU

Project No: 1340

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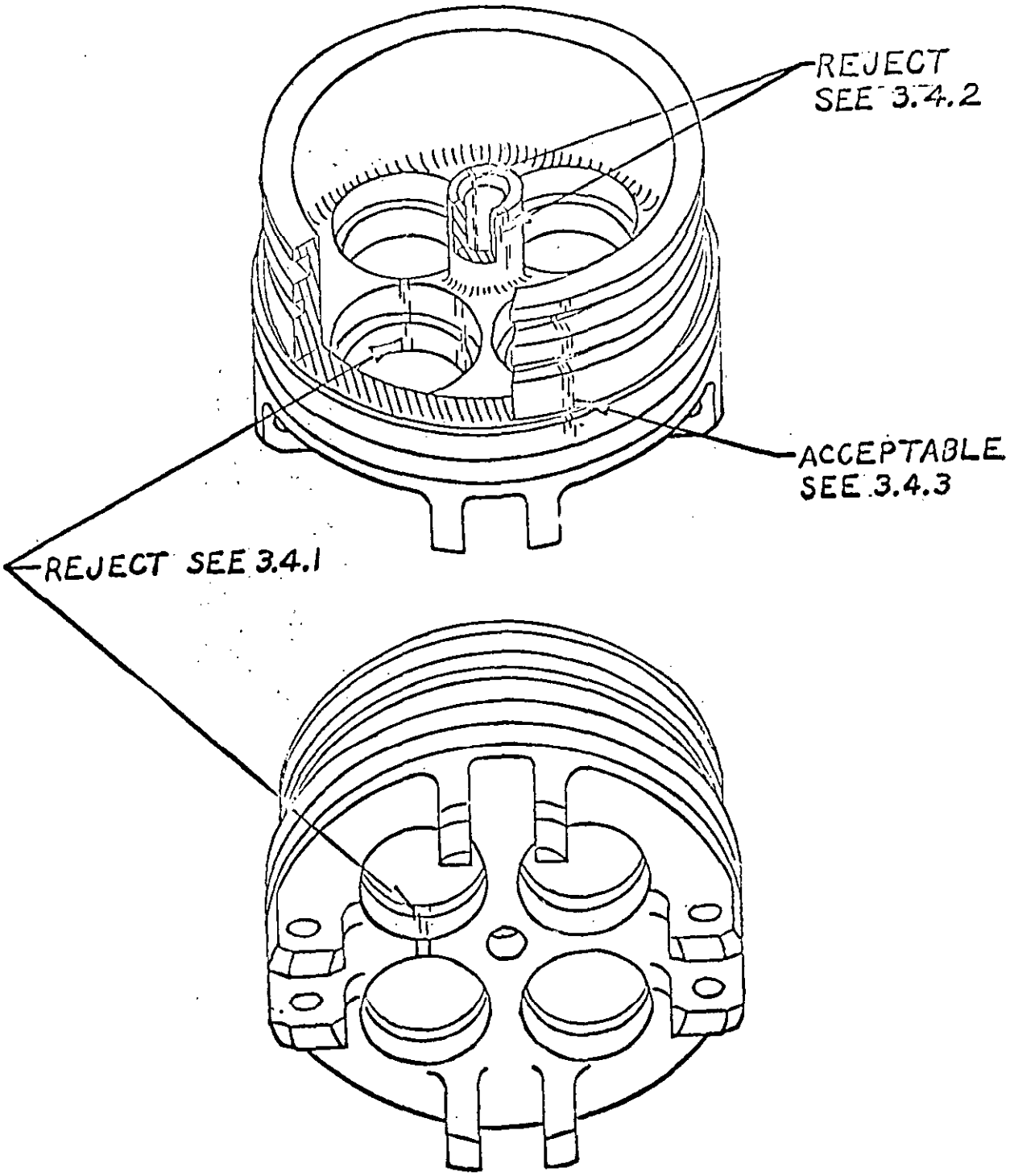


FIGURE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

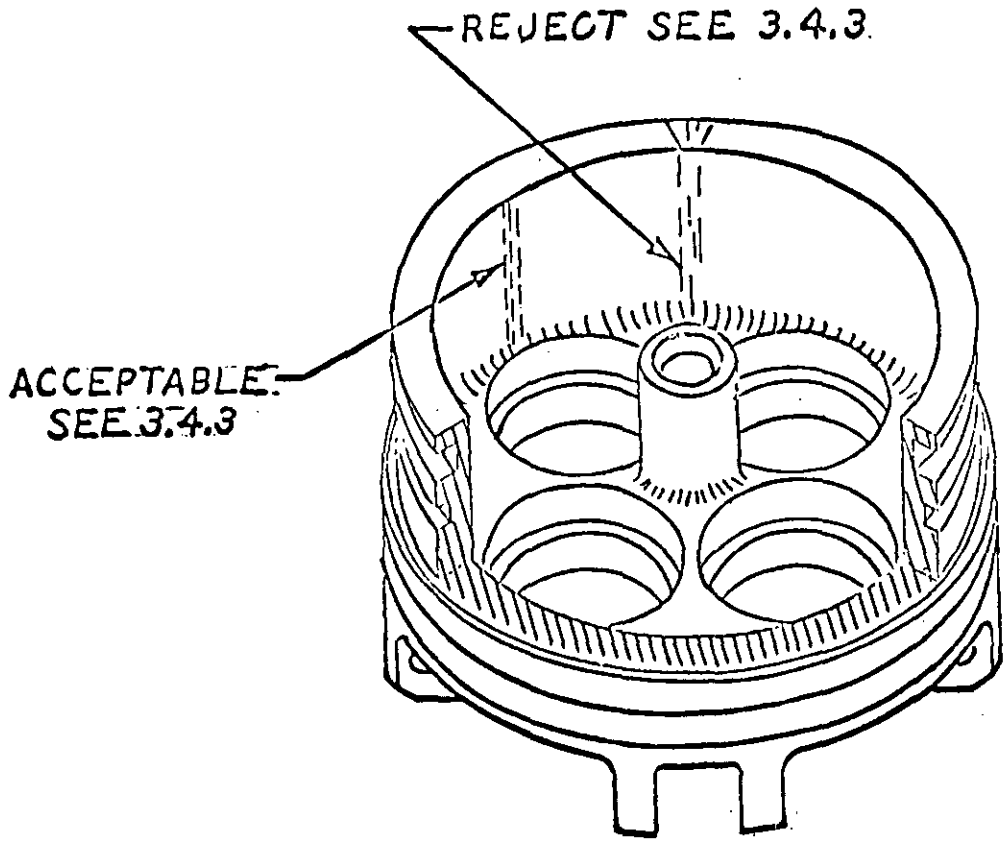


FIGURE 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
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